4.2.1 Algorithm Selection: Random Forest

A Random Forest Classifier was chosen. The model was trained on the engineered features to predict the Package category.

4.2.2 Model Training and Evaluation

The dataset was split into 80% for training and 20% for testing.

- **Training:** The RandomForestClassifier from scikit-learn was instantiated and trained. Hyperparameter tuning was performed using GridSearchCV to find the optimal number of trees and tree depth.

- **Evaluation:** The model's performance was evaluated on the test set.
  - **Accuracy:** Achieved an accuracy of **94%**.
  - **Confusion Matrix:** Revealed high precision and recall for all three package classes, indicating the model is not biased towards any single class.
  - **Feature Importance:** The model identified large_furniture_count and total_items as the most significant predictors.

4.3 Module 3: ML-Powered Dynamic Pricing and Offer System

4.3.1 Algorithm Selection: XGBoost

An XGBoost Regressor was used to predict the Price. This model can capture the non-linear relationship between features like time of day and price.

- **Training:** The model was trained using the same feature set, with Price as the target. The recommended package from the classification model was also included as a categorical feature.